

Do Gaussian Splats Keep the Nuclei?

An Object-Level Audit of Gaussian Volume Compression for Fluorescence Microscopy

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Abstract—Gaussian splats are being adopted to store and stream large 3D microscopy and medical volumes, with budgets chosen and quality reported by PSNR. We ask whether, at a given number of bytes on disk, standard analysis still finds the nuclei. On 12 densely annotated frames from two *C. elegans* embryos (2,958 labelled nuclei), we pair each of 80 fits of a Gaussian-splat compressor (Luxar) with JPEG2000 encoded to exactly the same bytes, and score four detectors against human labels: a blob detector, a classical 3D watershed, and Cellpose in its 2D-stitched and 3D modes. PSNR favours JPEG2000 on every frame and in all 80 of 80 pairs. The detectors disagree: below $\approx 300 \times$, 3D Cellpose finds more nuclei in splat reconstructions (+3.5 points, 10 of 12 frames), the watershed and 2D Cellpose find more in JPEG2000 (−4.0 and −9.4 points, all 12 frames), and the blob detector shows no consistent difference. The same network thus changes sides between its 2D and 3D modes. Beyond $\approx 300 \times$, at $\lesssim 3$ splats per nucleus, splats lose nuclei abruptly and every detector favours JPEG2000. PSNR is therefore not wrong but insufficient: it cannot tell which representation is safer for a given analysis. Gaussian volume compression should be evaluated at matched bytes, against a verified codec, with the detector that will be used.

I. Introduction

Light-sheet and confocal time-lapse microscopy produce terabytes per experiment, and medical archives face the same pressure. Representing a volume as a set of anisotropic 3D Gaussians [1] is attractive for such data: it is compact, continuous and renders directly on the GPU. Recent systems fit Gaussian splats to microscopy volumes at scale [2], to slice-based microscopy and ultrasound [3], to medical volumes for compact storage [4], [5], and to tomographic reconstruction [6], [7]. Quality is measured by PSNR or SSIM throughout, and budgets are chosen by PSNR: Luxar selects its splat count by self-supervised held-out PSNR [8] and describes its fits as intended for visualisation rather than quantitative analysis [2].

What a microscopist needs from a compressed volume is that the analysis still works. Whether a nucleus remains detectable hinges on a small, often faint peak. Radiology learned this for lossy codecs, where acceptability is judged by whether lesions remain detectable, not by PSNR [9]. Gaussian volume compression has not been held to that standard, nor compared with standard codecs at identical stored bytes.

Contributions. (i) A rate-detectability protocol: identical bytes on disk, dense human labels, four detectors,

frame-level tests with paired intervals over nuclei, and pass/fail rules committed before each new measurement, including a replication on eight further frames. (ii) PSNR picks the same winner at every size, but whether splats or a standard codec keep more nuclei depends on the detector—and the same network changes sides between its 2D and 3D modes. (iii) A budget cliff at $\lesssim 3$ splats per nucleus on every frame, which PSNR does not reveal. (iv) A baseline pitfall: a common JPEG2000 binding silently encodes a 3D array along the wrong axes, which had reversed our own high-ratio conclusion.

II. Related Work

Gaussian mixtures have long modelled nuclei for segmentation and lineage tracking [10]; our question concerns Gaussians as a storage format. Gaussian splatting [1] has been adapted to CT [7], [6], to slice-to-volume MRI with closed-form point-spread functions [11], and to volume compression [2], [3], [4], [5], each evaluated by image-fidelity metrics. JPEG2000 [12] and JPEG-XL [13] are the natural baselines at matched size. We use Cell Tracking Challenge labels [14], [15], a scale-normalised Laplacian of Gaussian [16], a distance-transform watershed, and Cellpose [17].

III. Methods

Data. Fluo-N3DH-CE [15]: *C. elegans* embryos imaged at $1.0 \times 0.09 \times 0.09 \mu\text{m}$ (8-bit, ≈ 12 MB per frame), every nucleus labelled. We use six frames per embryo (t110–t194 of sequence 01, t110–t185 of sequence 02; 102–362 nuclei each): four chosen first, eight added under rules fixed in advance. Embryo 1 t100, whose early nuclei are larger, is reported separately. Nuclei measure $3.6 \mu\text{m}$; all spatial parameters are fixed multiples of this diameter D .

Compressors. Luxar [2] with default settings, 1000 iterations and anisotropic voxel size, at seed budgets K from 500 to 6.4×10^4 ; its size is the .gsplats.zarr on disk, metadata included. Renders are mapped to raw intensities by a least-squares gain and offset (two numbers, not counted, which slightly favours Luxar). We also run Luxar’s own calibration—blind-spot held-out PSNR [8]—for its recommended K^* . JPEG2000 (OpenJPEG) codes each xy slice with a 2D wavelet, slices as components of one codestream, bisected to exactly the bytes of every

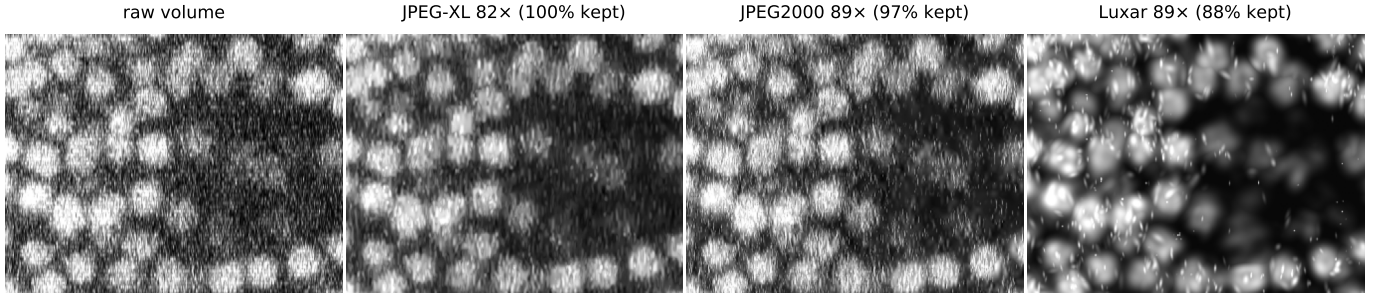


Fig. 1. The same region of embryo 1, t194 (3-slice maximum projection) from the raw volume, JPEG-XL, and JPEG2000 and Luxar at identical bytes; titles give whole-frame nuclei kept by 2D Cellpose. Both codecs preserve the textured appearance of nuclei; Luxar renders smooth blobs on a black background with thin streaks from elongated splats.

Luxar fit and to 50–800 KiB; JPEG-XL codes each slice, bisected to 50–800 KiB.

A baseline pitfall. Passing a (z, y, x) array to a common JPEG2000 binding encodes a z - y image with one component per x column, so the wavelet never runs along x . That configuration, which we used at first, costs ≈ 10 dB at $220\times$ and made splats appear far better than JPEG2000 beyond $200\times$; a regression test now checks the codestream header.

Detectors and scoring. (a) LoG: local maxima ranked by scale-normalised LoG response ($\sigma \in \{0.2, 0.27, 0.33, 0.4\}D$), top N kept, N = number of labels. (b) Watershed: background-subtracted, Otsu-thresholded foreground split by a distance-transform watershed in physical units, parameters fixed on raw volumes. (c) Cellpose 3 nuclei, 2D per slice stitched across z , and (d) the same model in its 3D mode (12 frames). On raw volumes they find 94%, 89%, 75% and 82% of the labelled nuclei. Detections match labels one-to-one within $0.6D$ (Table II varies it). Nuclei kept = nuclei found after compression / found in the raw volume by the same detector. Size-matched differences carry 95% bootstrap intervals over labelled nuclei, resampled jointly for raw and both reconstructions (Fig. 2); they are per pair and not corrected for multiple comparisons, so the claims rest on frame-level tests: frames as independent units, Wilcoxon signed-rank, Holm-corrected (Table I). Pass/fail rules were committed before each new measurement (rules, code and results: <https://github.com/asadrizvi64/DeepLearning-Seminar-TeamProject>); the two that failed are reported in narrowed form.

IV. Results

PSNR says JPEG2000; the detectors say it depends (Table I, Fig. 2). We treat the 12 frames as independent units and test the mean per-frame difference with Wilcoxon signed-rank tests, Holm-corrected over the eight detector-regime comparisons. PSNR is higher for JPEG2000 on every frame in both regimes and in all 80 of 80 pairs (by 1.0–6.8 dB). Below $300\times$, 3D Cellpose favours the splats (Luxar – JPEG2000 = +3.5 points of nuclei kept, 10 of 12 frames, $p_{\text{Holm}} = 0.010$), whereas the watershed and

TABLE I
Luxar minus JPEG2000 at identical bytes, frame level: mean over 12 frames of the per-frame mean difference (nuclei kept in points; PSNR in dB), 95% cluster-bootstrap interval over frames, and frames favouring Luxar: JPEG2000. Bold: Holm-corrected Wilcoxon $p < 0.05$.

Luxar – JPEG2000	below $300\times$		$300\times$ and beyond	
	mean [95% CI]	L : J	mean [95% CI]	L : J
PSNR (dB)	–2.4 [–2.7, –2.1]	0 : 12	–1.4 [–1.6, –1.3]	0 : 12
LoG	+1.3 [+0.0, +2.7]	8 : 4	–16.6 [–23.0, –10.2]	1 : 11
Watershed	–4.0 [–5.3, –2.7]	0 : 12	–19.3 [–23.8, –15.0]	0 : 12
Cellpose (2D)	–9.4 [–11.4, –7.3]	0 : 12	–23.2 [–28.0, –18.3]	0 : 12
Cellpose (3D)	+3.5 [+1.8, +5.5]	10 : 2	–18.2 [–23.8, –12.8]	0 : 12

2D Cellpose favour JPEG2000 (–4.0 and –9.4 points, every frame, $p_{\text{Holm}} = 0.004$ and 0.004). The blob detector shows no consistent difference (+1.3 points, 8 of 12 frames, $p_{\text{Holm}} = 0.110$): splats help it on embryo 2 (+2.4) but not embryo 1 (+0.3). Beyond $300\times$ (up to $603\times$) every detector favours JPEG2000 on 11–12 of 12 frames ($p_{\text{Holm}} \leq 0.004$), by –16.6 (LoG) to –23.2 (2D Cellpose) points; JPEG2000 still keeps 98–103% (LoG) and 96–100% (watershed) of nuclei there. PSNR is thus right at extreme ratios and for slice-wise segmentation, and wrong for 3D Cellpose at moderate ratios. JPEG-XL is nearly lossless up to its maximum distance (82 – $120\times$; worst case 91%).

Why the detector matters. Smooth splats can act as a denoiser for detectors that pool over a volume: under LoG and 3D Cellpose nuclei kept often exceeds 1 (Luxar at $45\times$ to $94\times$: 98–106% and 104–118%). Per-slice and threshold-based segmenters instead see smooth blobs on a black background with streaks (Fig. 1). The same Cellpose network shows this cleanly. At $45\times$ to $94\times$, JPEG-XL minus splats is +4 to +15 points of nuclei kept in 2D mode (intervals above zero on 4 of 4 frames) but –18 to –4 in 3D mode, where the 5-point penalty we tested for in advance appeared on 0 of 12 frames. The penalty is thus not a fixed property of the learned model, which supports an explanation involving processing geometry; switching modes also changes how predictions are combined, so it does not isolate the cause. The watershed sits in between (+0 to +5 points; under the pre-set 5 points on 11 of 12

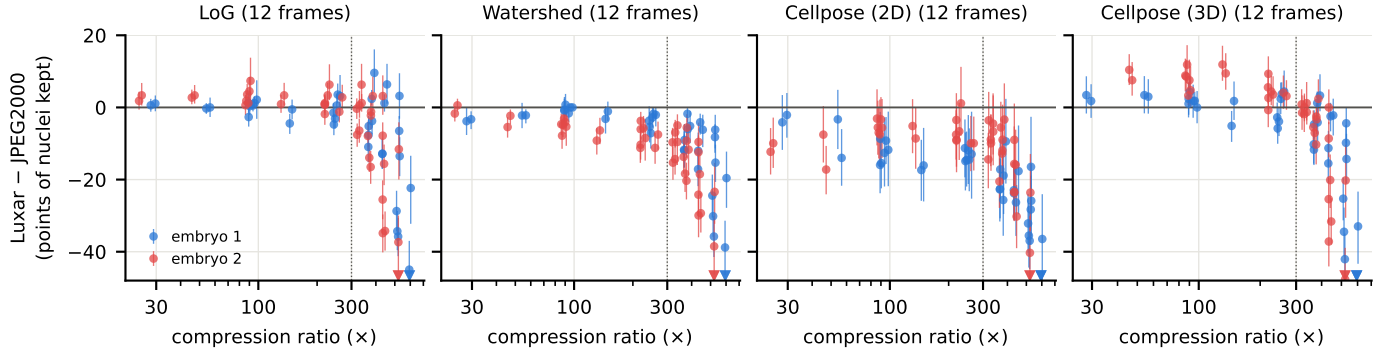


Fig. 2. Every Luxar fit against JPEG2000 at identical bytes: difference in nuclei kept (points; bars: 95% bootstrap intervals over nuclei) for four detectors. Above zero, splats keep more nuclei; triangles: below -48 points. Dotted line: $300\times$.

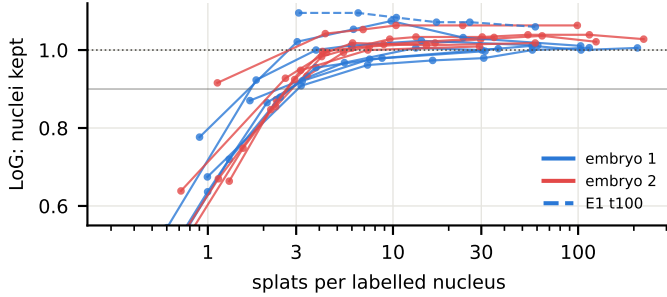


Fig. 3. The budget cliff: Log nuclei kept against splats per labelled nucleus, every Luxar fit on all frames (dashed: E1 t100).

frames).

The budget cliff and PSNR. Splats lose nuclei abruptly once they become scarce (Fig. 3): under LoG, nuclei kept crosses 90% at $1.1\text{--}3.1$ splats per labelled nucleus on all 12 frames, as pre-registered (≤ 3.5). Over budgets $K \geq 1000$, Luxar’s PSNR varies by at most 2.6 dB per frame (19.0–23.2 dB overall). Our pre-registered test that PSNR misses the cliff within $K = 1000\text{--}16\,000$ held on only 4 of 8 added frames: on smaller frames the cliff lies below that range. Survival—the fraction of the raw volume’s 100 strongest LoG detections still found, which needs no labels—tracks nuclei kept across fits (Spearman $\rho = 0.66$, $n = 80$) and can flag the cliff.

Luxar at its own budget. Luxar’s calibration recommends $K^* = 64,000$ on all 4 calibrated frames of both embryos—the top of our grid, held-out PSNR still rising. There, at $25\text{--}30\times$, it keeps 101–103% (LoG), 96–100% (watershed), 89–98% (2D) and 102–105% (3D Cellpose) of nuclei; JPEG2000 at the same bytes keeps 99–102%, in seconds rather than 17–46 min of fitting per volume on an A100 (5–40 min on an H100).

Reproducibility. Three independent Luxar fits of E1 t194 at each of three budgets differ by at most 2.1 (LoG), 3.5 (watershed) and 3.7 (2D Cellpose) points; Cellpose’s own scores move by up to 8 points between neighbouring codec sizes. On E1 t100, with larger early nuclei, the blob detector favours splats more strongly (8–11 points at 159--

TABLE II
Match-radius sensitivity (four first frames): nuclei found in the raw volumes and nuclei kept near $90\times$ (mean).

Radius	LoG			Cellpose (2D)		
	raw	JXL	Luxar	raw	JXL	Luxar
$0.4D$	1009	1.00	1.02	658	1.01	0.84
$0.6D$	1032	1.00	1.01	788	0.97	0.88
$0.8D$	1034	1.00	1.01	796	0.99	0.90

$537\times$).

V. Discussion and Limitations

On this data no storage format is best for every analysis. JPEG-XL is nearly lossless to $\approx 100\times$; beyond it, JPEG2000 kept more nuclei for slice-wise and threshold-based segmentation, and for every detector we tested past $300\times$, while Gaussian splats helped volumetric Cellpose at moderate ratios, at the cost of minutes of GPU fitting. None of this establishes that any format is safe in general. PSNR is not random—its preference for JPEG2000 matched most of these outcomes—but it ranks the two formats the same way at every size, so it is insufficient to determine detector-specific outcomes. We recommend comparing Gaussian volume compression with a verified codec at identical bytes, reporting object-level recall with the detector that will be used downstream, and checking a splat budget at the object level before trusting it (survival is a label-free start).

Limitations: one organism (two embryos, 12 frames); one Gaussian compressor at default settings (tuned or entropy-coded splats may fare better); a 2D-per-slice JPEG2000 (a 3D wavelet would likely be stronger still); frames, not embryos, as the units of replication (two embryos); and a fixed D that suits early, larger nuclei (t100) less well. The same audit for small low-contrast lesions in CT is future work.

VI. Conclusion

Measured at identical bytes against human labels, Gaussian-splat compression keeps nuclei well at its own

budget, gains or loses nuclei against JPEG2000 at moderate ratios depending on the detector—even between the 2D and 3D modes of one network—and loses them abruptly below a budget that PSNR cannot identify. Object-level, size-matched, detector-specific evaluation should accompany Gaussian volume compression.

References

- [1] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, “3D gaussian splatting for real-time radiance field rendering,” *ACM Transactions on Graphics*, vol. 42, no. 4, 2023.
- [2] L. A. Royer, “Luxar: Gaussian splatting for microscopy and scalable interactive web visualisation of multidimensional scientific data,” Zenodo, <https://zenodo.org/records/22912050>, 2026, preprint; software at <https://github.com/royerlab/luxar>.
- [3] D. Kong, Y. Wang, W. Guo, Y. Bu, B. Zhang, Y. Duan, X. Yue, W. Du, Y. Zhong, Y. Chen, and C. Ma, “GaussianPile: A unified sparse Gaussian splatting framework for slice-based volumetric reconstruction,” *arXiv:2603.20611*, 2026.
- [4] X. Li and Q. Gu, “GSToken: Geometry-structured Gaussian tokens for compact 3D medical image representation,” *arXiv:2608.17425*, 2026.
- [5] M. R. Martin, J. Insley, V. A. Mateevitsi, S. Rizzi, and K.-L. Ma, “Fixed-budget Gaussian volume encoding with structure-aware allocation,” *arXiv:2608.14112*, 2026.
- [6] Y. Li, X. Fu, H. Li, S. Zhao, R. Jin, and S. K. Zhou, “3DGR-CT: Sparse-view CT reconstruction with a 3D Gaussian representation,” *Medical Image Analysis*, vol. 103, p. 103585, 2025.
- [7] R. Zha, T. J. Lin, Y. Cai, J. Cao, Y. Zhang, and H. Li, “R²-Gaussian: Rectifying radiative Gaussian splatting for tomographic reconstruction,” in *Advances in Neural Information Processing Systems*, 2024.
- [8] J. Batson and L. Royer, “Noise2Self: Blind denoising by self-supervision,” in *Proceedings of the 36th International Conference on Machine Learning*, 2019, pp. 524–533.
- [9] D. A. Koff and H. Shulman, “An overview of digital compression of medical images: can we use lossy image compression in radiology?” *Canadian Association of Radiologists Journal*, vol. 57, no. 4, pp. 211–217, 2006.
- [10] F. Amat, W. Lemon, D. P. Mossing, K. McDole, Y. Wan, K. Branson, E. W. Myers, and P. J. Keller, “Fast, accurate reconstruction of cell lineages from large-scale fluorescence microscopy data,” *Nature Methods*, vol. 11, no. 9, pp. 951–958, 2014.
- [11] M. Dannecker, S. Jia, N. Stolt-Ansó, N. Girard, G. Auzias, F. Rousseau, and D. Rueckert, “Fast and explicit: Slice-to-volume reconstruction via 3D Gaussian primitives with analytic point spread function modeling,” in *Medical Imaging with Deep Learning (MIDL)*, 2026.
- [12] D. S. Taubman and M. W. Marcellin, *JPEG2000: Image Compression Fundamentals, Standards and Practice*. Kluwer Academic Publishers, 2002.
- [13] J. Alakuijala et al., “JPEG XL next-generation image compression architecture and coding tools,” in *Applications of Digital Image Processing XLII*, Proc. SPIE 11137, 2019.
- [14] V. Ulman et al., “An objective comparison of cell-tracking algorithms,” *Nature Methods*, vol. 14, pp. 1141–1152, 2017.
- [15] M. Maška et al., “The cell tracking challenge: 10 years of objective benchmarking,” *Nature Methods*, vol. 20, pp. 1010–1020, 2023.
- [16] T. Lindeberg, “Feature detection with automatic scale selection,” *International Journal of Computer Vision*, vol. 30, no. 2, pp. 79–116, 1998.
- [17] C. Stringer, T. Wang, M. Michaelos, and M. Pachitariu, “Cellpose: a generalist algorithm for cellular segmentation,” *Nature Methods*, vol. 18, pp. 100–106, 2021.